

Cole Campbell

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SUMMARY

New-grad data scientist and ML engineer with a portfolio in generative AI, deep learning, and statistical modeling. Builds systems end to end, from data pipelines through model integration in production backends. Currently pursuing an M.S. in Data Science at Arizona State University on top of a 2026 B.S., and looking for an entry-level role in data and AI.

EDUCATION

Arizona State University

B.S. in Data Science (Concentration in Computer Science), 05/2026

GPA: 3.52

Arizona State University

M.S. in Data Science, Analytics, and Engineering (Computational Mathematics & Data), Expected 05/2027

TECHNICAL SKILLS

Languages:	Python, R, Dart, C/C++, JavaScript (Node.js), SQL, Bash, MATLAB
ML / AI:	Deep learning (CNNs, RNNs/LSTMs, Transformers), generative models (Diffusion, GANs, StyleGAN), LLMs & prompt engineering, LoRA/QLoRA/PEFT, transfer learning, neural style transfer, audio/music generation, model evaluation (FID, CLIP, LPIPS, BLEU, perplexity)
Frameworks:	PyTorch (torchvision/torchaudio), TensorFlow, Hugging Face (Transformers, Diffusers, Accelerate), scikit-learn, NumPy, Pandas, SciPy, StatsModels, FastAPI, Flutter/Riverpod
Data Science:	Regression & statistical modeling, classification, time series, Bayesian, clustering, longitudinal/panel analysis, feature engineering, A/B testing, causal inference, hypothesis testing, visualization (Matplotlib, Seaborn, Plotly)
Data Engineering:	ETL & data pipelines, REST API integration, SQLAlchemy + Alembic, PostgreSQL, Firebase, Docker, CUDA, cloud (AWS, GCP, Azure ML)
Tools:	Git, Linux, Jupyter, VS Code, OpenCV, librosa, ParaView/VTK, pytest, Weights & Biases, Kaggle/NBA APIs, REST/OpenAPI
Math:	Linear algebra, probability, optimization, numerical methods, stochastic processes, statistical theory

PROJECTS

Multi-Architecture Generative Content Studio — Deep Learning Capstone

PyTorch • Transformers • Diffusion • GANs • CNNs • RNNs

- Built a unified generative system producing story text, scene illustrations, character portraits, and background music from a single prompt.
- Integrated GPT-2, Stable Diffusion, StyleGAN2/3, VGG16/19, and LSTM into one cohesive pipeline.
- Fine-tuned GPT-2 for narrative generation, entity extraction, and structured scene descriptions; implemented Stable Diffusion with LoRA + ControlNet for composition-controlled scenes.

Tempo — AI Scheduling Assistant

FastAPI • Flutter • PostgreSQL • LLM / NLP • Constraint Optimization

- Built a cross-platform scheduling app (Flutter front end, FastAPI back end) that generates alternative day/week plans, compares them, and applies the one the user picks.
- Designed a constraint-based scheduling engine that orders tasks, builds the daily timeline, resolves conflicts, and merges blocks into a coherent schedule.
- Engineered a scenario pipeline supporting generation, diff-based comparison, trade-off analysis, and undo/redo state management.

PSID Panel Dataset Analysis — Longitudinal Economics Project

Python • Statistical Modeling • Longitudinal Analysis

- Analyzed a 17,000-observation PSID panel (1999–2023) to study income and wealth dynamics.
- Built the full pipeline: cleaning, transformation, feature engineering, regression modeling, and visualization.
- Found wealth roughly twice as unequal as income (wealth Gini ~0.85 vs income Gini ~0.45), with the median wealth-to-income ratio rising from 1.25 to 2.05 across the panel.

Portfolio & demos: github.com/cjcamp16

EXPERIENCE

Handshake

AI Trainer — May 2026 – Present

- Trained AI models based on their ability to reason and evaluate tasks across mathematics, codebases, and visualizations.